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Hike explores park's geological history

by [Jason Klaiber](#) — September 9, 2024 in [Eagle Bulletin](#), [Outdoors](#)

A hike at Clark Reservation led by geologist Ashley Prow took a look at the Jamesville park's rocks and fossils. (Jason Klaiber)

TOWN OF DEWITT – A group of hikers started off the month of September by learning about the rocks, fossils, formations and geological history of Clark Reservation State Park.

The walk, led by geologist Ashley Prow on Sept. 1, took the loop from the Table Rock trail to Dry Lake, and it primarily focused on sedimentary rocks that were deposited about 380 million years ago.

Prow said that due to plate tectonics, New York State was located at the equator during the middle part of the Devonian Period.

That reef system is known as the Onondaga Formation, which comprises the upper limestone bluffs of the Clark Reservation cataract. That reef system looks different than it once did, and its corals are all extinct now, Prow said, but they have similar relatives in the modern world.

“Fast forward approximately 12,000 years ago during the last ice age, and New York would’ve at that point moved to its current position at about 40 degrees latitude,” Prow said.

She added that because it was an ice age it was obviously very cold—colder than the temperatures Central New Yorkers are used to—and the state park in Jamesville would’ve been underneath approximately two kilometers of glaciers, which would’ve exceeded the height of the Empire State Building.

“These glaciers would have carved out the terrain and then also compressed the sediment beneath it,” Prow said. “During the last 10,000 years or so, these glaciers then would have retreated, and then all of that melt water would’ve come rushing through the ravines that the glacier would’ve gauged out.”

Prow said it’s difficult for anyone now to imagine the volume of water that was flowing over the area where she and the other hikers were standing as a result of the sudden removal of all that glacial weight, but she said it’s estimated it was over two times that of Niagara Falls’ present-day volume.

That forceful current of water led to the movement of the sediment in the direction of the flow of the water and a series of fissures that uplifted the ground and cracked and sculpted the brittle bedrock, creating ripple marks that can still be pointed out. Rain then percolated through and progressively expanded those cracks even more to produce a karstic landscape made up of dissolved limestone and sinkholes.

Prow also said that anywhere from 7,000 to 12,000 years ago the overlook near the nature center that provides a view of the meromictic lake below was itself a waterfall, making the lake its plunge basin. That waterfall then gradually eroded into the bedrock as part of a process referred to as “undercutting,” she went on to say.

The guided hike later included some looks at fossils in their natural burial positions. Prow said fossilization requires “very specific conditions” to occur; for one, the sediment has to be soft and fine-grained, plus the burial of the fossil has to happen almost instantaneously.

At the different stops along the way, fossil finds included scattered fragments of horn corals, which are called that due to their shape, as well as the insect-related arthropods similar in ways to horseshoe crabs called trilobites and the variety of sponge known as stromatoporoids, the most primitive form of animal.

Another fossil shown to the group was the remnant of a crinoid, which are also called sea lillies. Those specimens are related to starfish and sea urchins, and because of their disc shape people used to think in ancient times that they were currency for fairies.



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